Class 11 Biology— Drug Addiction & Its Effects

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1. Why do some drugs make a person addicted to them?

Answer: Some drugs affect the **reward system of the brain** and produce feelings of pleasure, excitement, relaxation, or euphoria (intense happiness).

Many addictive drugs increase the activity of a neurotransmitter called **dopamine** in the brain.

How addiction develops

Drug taken → Brain reward system activated → Dopamine activity increases → Feeling of pleasure → Person wants to repeat the experience

With repeated use:

- **Tolerance develops:** The same amount of drug produces less effect, so the person may need a larger dose.
- **Dependence develops:** The brain and body begin to function abnormally without the drug.
- **Craving develops:** The person develops a strong desire to take the drug again.
- **Withdrawal symptoms may occur:** Stopping the drug can cause anxiety, restlessness, sweating, nausea, sleep problems, etc.

One-line viva answer

Addictive drugs alter the brain's reward system, particularly dopamine signalling, causing craving, tolerance and dependence.

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2. How do some drugs attach to neurons and stop normal nerve impulses?

The question in the notebook can be scientifically stated as:

How do drugs interfere with neurotransmission at a synapse?

First understand normal transmission.

Normal nerve impulse

Neuron 1 → Neurotransmitter released → Receptor on Neuron 2 → Signal continues

Neurons usually do **not physically touch each other**. A small gap called the **synaptic cleft** is present between them.

When an electrical impulse reaches the end of a neuron:

1. **Neurotransmitters are released** from the presynaptic neuron.
2. They cross the **synaptic cleft**.
3. They bind to specific **receptor proteins** on the next cell.

4. This changes the activity of the next cell and may generate or inhibit another nerve impulse.

What drugs can do

Different drugs interfere with this system in different ways.

Some drugs may:

- **Mimic neurotransmitters** and activate their receptors.
- **Block receptors**, preventing the normal neurotransmitter from binding.
- **Prevent reuptake** of neurotransmitters, leaving unusually high amounts in the synapse.
- **Increase or decrease neurotransmitter release.**
- **Alter ion channels**, changing the electrical activity of neurons.

Therefore, saying that every drug simply "attaches to the neuron and stops another protein" is **not quite correct**. Different drugs have different mechanisms.

One-line viva answer

Drugs can alter nerve transmission by acting on neurotransmitter receptors, neurotransmitter release or reuptake, or ion channels at neurons and synapses.

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3. Types of addictive drugs and their effects

For Class 11, remember the major groups rather than trying to memorise a huge list.

Type	Examples	Main effect
Opioids	Heroin, morphine	Depress CNS activity and produce pain relief/euphoria; overdose can suppress breathing
Stimulants	Cocaine, amphetamines	Increase alertness and CNS activity; can raise heart rate and blood pressure
Depressants / sedatives	Some sedative-hypnotic drugs	Slow CNS activity; may cause drowsiness and impaired coordination
Cannabinoids	Marijuana/ganja, hashish	Affect perception, mood, memory and coordination
Hallucinogens	LSD	Alter perception, thoughts and sensory experiences

Important NCERT-related terms

A. Opioids

Opioids bind to **opioid receptors** present in the central nervous system and other tissues.

Heroin is an opioid drug.

Effects may include:

- Euphoria
- Drowsiness
- Reduced pain sensation
- Slowing of body functions
- Respiratory depression in overdose
- Dependence and addiction

B. Cannabinoids

Cannabinoids interact with **cannabinoid receptors** in the body, especially in the brain.

Examples include preparations of **Cannabis sativa**, such as marijuana/ganja and hashish.

They can affect:

- Memory
- Coordination
- Perception
- Mood
- Reaction time

C. Cocaine

Cocaine is a powerful **central nervous system stimulant**.

It strongly affects dopamine signalling, particularly by **blocking dopamine reuptake**, so dopamine remains active in the synapse for longer.

This can cause:

- Increased alertness
- Euphoria
- Increased heart rate
- Increased blood pressure
- Restlessness

- Anxiety
- High addiction potential

Large doses can produce serious cardiovascular and neurological complications.

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4. How does cocaine affect cardiovascular health?

This is especially important because it is specifically written in your practical notebook.

Cocaine stimulates the sympathetic nervous system.

It increases the effects of neurotransmitters such as **dopamine and noradrenaline (norepinephrine)**.

Cardiovascular effects

1. Increased heart rate (tachycardia):
The heart begins beating faster.

2. Increased blood pressure:
Cocaine causes strong sympathetic stimulation.

3. Vasoconstriction:
Blood vessels become narrower.

4. Increased workload on the heart:
The heart requires more oxygen because it is beating faster and harder.

5. Reduced oxygen supply to heart muscle:
Constriction of coronary arteries can reduce blood flow to the heart.

6. Abnormal heart rhythm (arrhythmia):
Cocaine can interfere with normal electrical activity of the heart.

7. Increased risk of heart attack:
The combination of increased oxygen demand and reduced blood supply can damage cardiac muscle.

Easy mechanism to remember

Cocaine → Sympathetic stimulation → ↑ Heart rate + ↑ Blood pressure + Vasoconstriction → Greater stress on heart → Risk of arrhythmia/heart attack

Viva answer

Cocaine is a powerful stimulant. It increases heart rate and blood pressure and causes vasoconstriction. This increases the workload of the heart while potentially reducing its blood supply, increasing the risk of arrhythmia and heart attack.

Important Terms to Learn

Addiction: A condition involving compulsive drug use despite harmful consequences.

Dependence: A state in which the body or brain has adapted to a drug and stopping it can produce withdrawal symptoms.

Tolerance: A condition in which a person needs a larger amount of a drug to obtain an effect similar to what a smaller amount previously produced.

Withdrawal: Physical or psychological symptoms that can appear when a dependent person stops or substantially reduces the drug.

Neurotransmitter: A chemical messenger used by neurons to communicate with other cells.

Receptor: A protein that recognizes and responds to a particular chemical signal.

Synapse: The junction through which one neuron communicates with another cell.

Dopamine: A neurotransmitter involved in movement, motivation, reinforcement and reward.

Very Likely Practical Viva Questions

Q1. What is drug addiction?

Ans. Drug addiction involves compulsive use of a drug despite harmful consequences and is associated with changes in brain circuits involved in reward, motivation and self-control.

Q2. What is a neurotransmitter?

Ans. A neurotransmitter is a chemical messenger released by neurons to communicate with other cells.

Q3. What is a synapse?

Ans. A synapse is the junction through which a neuron communicates with another neuron or target cell.

Q4. Which neurotransmitter is strongly associated with the brain's reward pathway?

Ans. Dopamine is strongly involved in reward and reinforcement.

Q5. What type of drug is cocaine?

Ans. Cocaine is a powerful **central nervous system stimulant**.

Q6. How does cocaine affect dopamine?

Ans. Cocaine blocks dopamine reuptake, causing dopamine to remain in the synapse for longer and increasing dopamine signalling.

Q7. What is tolerance?

Ans. Tolerance means that repeated use reduces the effect of a given dose, so a larger amount may be required to produce a similar effect.

Q8. What is withdrawal?

Ans. Withdrawal refers to symptoms that can occur when a dependent person stops or greatly reduces drug use.

Q9. What type of drug is heroin?

Ans. Heroin is an **opioid**.

Q10. Where do opioids act?

Ans. They bind to **opioid receptors**, especially in the central nervous system.

Q11. What are cannabinoids?

Ans. Cannabinoids are compounds that act on cannabinoid receptors and can alter mood, memory, perception and coordination.

Q12. Why is cocaine dangerous for the heart?

Ans. It can increase heart rate and blood pressure, constrict blood vessels and disturb heart rhythm, increasing cardiovascular risk.

Quick Memorise

1. Why are drugs addictive?
They can alter the brain's reward system, especially dopamine signalling. Repeated use can produce craving, tolerance and dependence.

2. How do drugs interfere with neurons?
They can mimic or block neurotransmitters, alter their release or reuptake, or affect receptors and ion channels, thereby changing normal nerve signalling.

3. Main addictive drug groups?
Remember **opioids, stimulants, cannabinoids, depressants and hallucinogens**, along with one example and their major effects.

4. Cocaine and the heart?
Cocaine → sympathetic stimulation → increased heart rate + increased BP + vasoconstriction → increased cardiac workload → risk of arrhythmia and heart attack.